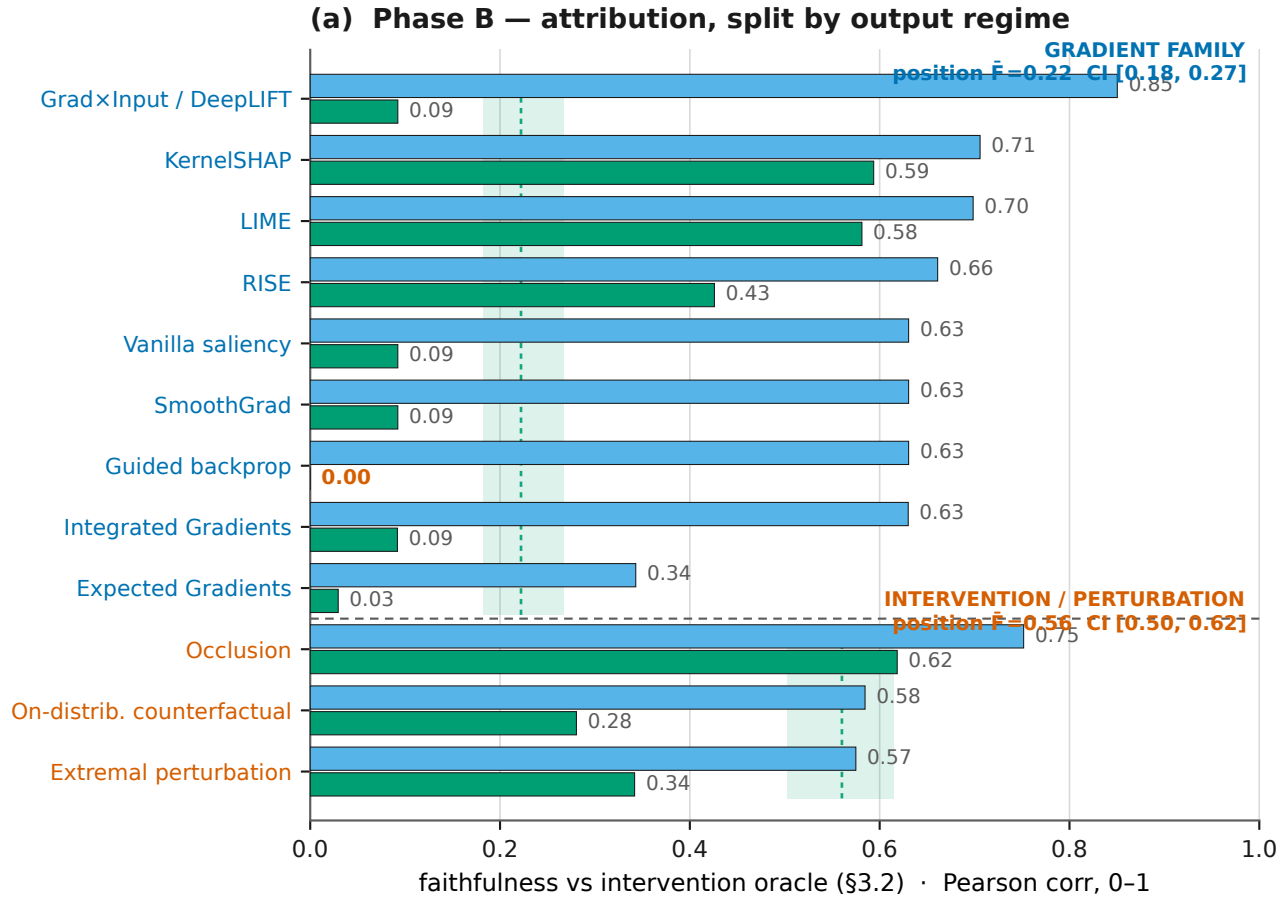
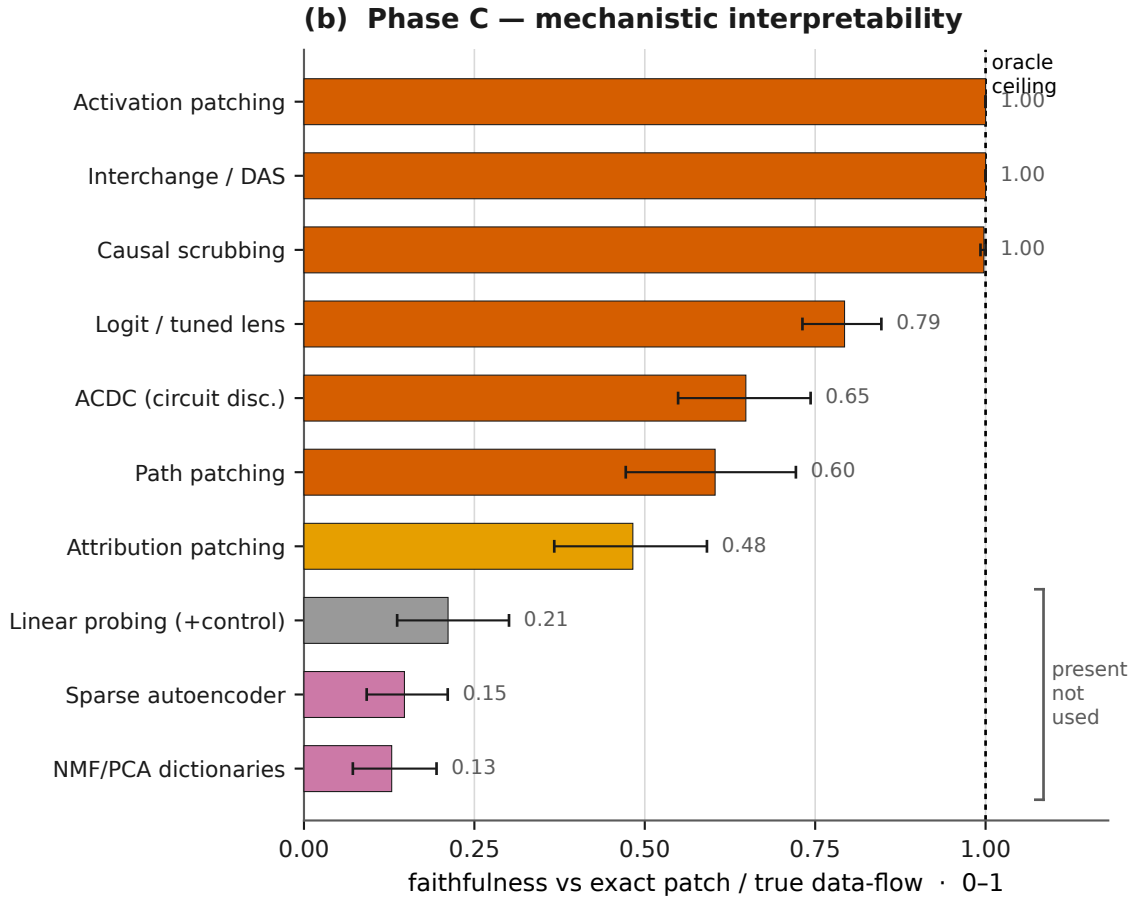
Attribution (Phase B) vs mechanistic interpretability (Phase C): faithfulness on the bit-exact VCS

The 22 Phase-B + Phase-C method rows (a subset of the 30 methods + the oracle = 31-row leaderboard; the cross-tradition scatter is Fig. 2).

HEADLINE — across ALL output regimes, intervention/causal methods are more faithful to the true mechanism than gradient/correlational ones. Causal/intervention $\bar{F}$=0.709 (95% CI [0.681, 0.736]) vs gradient/correlational $\bar{F}$=0.384 (95% CI [0.356, 0.410]) — a 0.324 gap, 95% CI [0.297, 0.351] (excludes 0: robust). On the POSITION/INDEX regime alone (discrete sprite-position outputs), the gap is now SIGNIFICANT on the 44-game battery: causal 0.560 vs gradient 0.222; family-mean gap 0.338, 95% CI [0.293, 0.383] over 44 games (EXCLUDES 0).





Data: committed records (leaderboard + bootstrap CIs + position bootstrap); 44-game scored battery, 30-frame horizon; pure read — no experiment re-run. Exact paths & per-number provenance in the Supplement.